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Mechanistic study of the competition between carbon dioxide reduction and hydrogen evolution reaction and selectivity tuning *via* loading single-atom catalysts on graphitic carbon nitride†

Joel Jie Foo,^{a,b} Sue-Faye Ng,^{a,b} Mo Xiong^{*c} and Wee-Jun Ong ^{*a,b,d,e,f}

In the context of catalytic CO₂ reduction (CO₂RR), the interference of the inherent hydrogen evolution reaction (HER) and the possible selectivity towards CO have posed a significant challenge to the generation of formic acid. To address this hurdle, in this work, we have investigated the impact of different single-atom metal catalysts on tuning selectivity by employing density functional theory (DFT) calculations to scrutinize the reaction pathways. Single-atom catalysts supported on carbon-based systems have proven to be pivotal in altering both the activity and selectivity of the CO₂RR. In this study, a series of single-atom-metal-loaded g-C₃N₄ monolayers (MCN, M = Ni, Cu, Zn, Ga, Cd, In, Sn, Pb, Ag, Au, Bi, Pd and Pt) were systematically examined. Through detailed DFT calculations, we explored their influence on reaction selectivity between the *COOH and *OCHO intermediates. Notably, NiCN favors the reaction *via* the *OCHO route, with a significantly lower rate-determining potential of 0.36 eV, which is approximately 73.5% lower than that of the CN system (1.36 eV). Most importantly, the Ni single-atom catalyst with lower coordination significantly enhances CO₂ adsorption, promoting CO₂RR over HER. Overall, this study, guided by DFT calculations, provides a theoretical prediction of how the selection of single-atom metal catalysts can effectively modulate the reaction pathway, thereby offering a potential solution for achieving high product selectivity in CO₂RR.

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1. Introduction

The development of green and sustainable energy is particularly important in the contemporary era to address devastating environmental issues such as the ever-increasing atmospheric CO₂ concentration due to emissions from the combustion of the depleting fossil fuel by the energy sector. In fact, CO₂ emission is the main reason that drives global climate change towards an irreparable state, gradually reshaping ecosystems to uninhabitable environments.^{1–3} Thus, technologies to reduce

CO₂ emissions and development of alternative green and sustainable fuels are urgently needed to address the deteriorating environmental conditions. Tremendous research efforts have been devoted to capturing and containing greenhouse gases, such as carbon capture and reformation strategies and catalysis technology. However, the conventional hydrogenation of CO₂ is highly detrimental from the environmental perspective due to the carbon emission from this thermally activated reaction. Besides, the conventional reaction requires precious metals such as Rh and Ru as the catalyst, thus lowering the profitability of the technology.^{4,5} Whereas, sustainable catalytic technologies such as electrocatalysis and photocatalysis that convert CO₂ into value-added fuels (such as CO,^{6–9} HCOOH,^{10–15} CH₃OH^{16–18} and even C₂ products^{19–23}) are ideal solutions to reduce the atmospheric CO₂ concentration and simultaneously generate sustainable fuels using CO₂ as the raw material to replenish the depletion of the fuel, killing two birds with one stone.^{24–26} Among carbonaceous products, formic acid/formate has received considerable attention due to its potential to serve as a green energy carrier. The high density of formic acid (1.22 g mL^{−1}) endows it with a considerable volumetric H₂ capacity (53 g L^{−1}), which boosts its potential as a liquid H₂ carrier for hydrogen fuel-cell vehicles.²⁷

^aSchool of Energy and Chemical Engineering, Xiamen University Malaysia, Sepang, Selangor Darul Ehsan, 43900, Malaysia. E-mail: weejun.ong@xmu.edu.my

^bCenter of Excellence for NaNo Energy & Catalysis Technology (CONNECT), Xiamen University Malaysia, Sepang, Selangor Darul Ehsan, 43900, Malaysia

^cMOE Key Laboratory for Non-Equilibrium Synthesis and Modulation of Condensed Matter, School of Physics, Xi'an Jiaotong University, Xi'an 710049, Shanxi, China. E-mail: xiongmo@xjtu.edu.cn

^dState Key Laboratory of Physical Chemistry of Solid Surfaces, College of Chemistry and Chemical Engineering, Xiamen University, Xiamen 361005, China

^eGulei Innovation Institute, Xiamen University, Zhangzhou 363200, China

^fShenzhen Research Institute of Xiamen University, Shenzhen, 518057, China

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In addition, formic acid is also one of the significant intermediates formed during the synthesis of oxygen-containing compounds such as alcohol, ester and acids in the field of syngas production.^{28,29}

Yet, the industrial application of catalytic CO₂ reduction reactions (CO₂RR) is greatly restrained by the inert and linear geometry of CO₂, which requires -1.9 V vs. NHE for CO₂RR activation.³⁰ The second challenge is the multiple-electron kinetics, which makes it less competitive to the hydrogen evolution reaction (HER), which only requires two electrons.³¹ Thirdly, the complex reaction pathway and low product selectivity can lead to endless possibilities of carbonaceous products.^{32,33} Thus, the selectivity of the reaction to specific carbon products is highly dependent on the interactions between the catalyst and the key intermediate, which is governed by the energy barrier of the transition state evolution step in each of the proton-coupled electron transfer processes.

Graphitic carbon nitride (g-C₃N₄) is an emerging material in the field of catalysis owing to its extraordinary electrical conductivity and low internal resistance, which allows the generated charge carriers to spontaneously transmit to the active sites.^{34–36} The high thermal and chemical stability demonstrate the robust stability of the catalyst under harsh reaction environments. Its low toxicity ensures safer handling and operation while lowering the destruction risk of the ecosystem. Most importantly, its semiconducting properties, with a band gap of 2.7 eV,³⁷ allow photon harvesting in the visible-light region (which accounts for roughly 43% of the solar spectrum),³⁸ signifying that it can potentially serve as a photocatalyst.

The recent advent of nanomaterials with engineered shapes and dimensions presents promising modification strategies for advancement in the realm of catalysis. Among various classes of nanoarchitectures, 2D g-C₃N₄ has emerged as one of the prominent materials owing to its intrinsic properties that significantly enhance the catalytic performance in CO₂RR. (1) The 2D nanostructure endows ample active sites for the adsorption and activation of the CO₂ reactant. It eases the modification strategies, such as defect engineering and heterojunction engineering with another catalyst, by presenting highly exposed surface atoms, which can be removed or substituted.^{35,39} (2) 2D materials generally have a less light-shielding effect and hence can maximize the light absorption ability.⁴⁰ (3) The short diffusion distance of the generated charge carrier to travel to the active sites reduces charge recombination before reaching the active sites.^{40,41} Despite the intrinsic advantages of 2D g-C₃N₄, carbon nitride is unable to deliver satisfactory catalytic CO₂RR performance owing to the poor activation of adsorbed CO₂.

Recently, there has been a surge in the investigation of single-atom catalysts (SAC) for maximum atomic utilization to lower the cost and simultaneously overcome the large energy barrier of the first protonation steps of the CO₂RR.^{42–44} The introduction of foreign atoms was found to alter the reaction pathway with a lowered energy barrier and redistribute the electrons on the surface of the catalyst.^{45,46} He *et al.* found

that the addition of Pd SAC enhances CO₂ to CO conversion, with the Pd-N₄ moiety sites leading to the *HOCO key intermediate reaction pathway.⁴⁷ Liu *et al.* reported that the loading of Co SAC on the graphdiyne system induced charge redistribution; the Co SAC induced a negative net charge of $-3.76e$, and the neighboring C atoms exhibited positive values of around $0.5\text{--}0.7e$, indicating that the charge transfer from the substrate contributes to the adsorption and activation of CO₂ molecules.⁴⁸ As such, graphitic carbon nitride with a tri-s-triazine structure has a large interstitial void surrounded by 6 unsaturated nitrogen atoms, which represents a potential anchoring site for SAC.^{49,50} Although machine learning and computational studies of SAC-loaded g-C₃N₄ have been done previously,^{51–58} these studies involve either non-CO₂RR applications or the metal selection used for calculations is only limited to one or two SAC(s). Therefore, there is a lack of understanding of the effect of different metal SAC (M-SAC) on the underlying reaction mechanism and selectivity of the complex CO₂RR toward the formation of formic acid.

By leveraging the idea of combining SAC and 2D g-C₃N₄, in this work, we investigated the effect of different M-SACs on the two-proton-coupled CO₂RR electron transfer pathway that leads to the formation of formic acid. The formation energy calculation served as the first layer of screening to evaluate the feasibility of the system before the reaction, followed by the evaluation of CO₂RR selectivity between HCOOH and CO, HER competition, density of states and Bader charge analysis to unravel the effects of various metals on the product selectivity of the CO₂RR *via* density functional theory (DFT) calculations. Among the studied SACs, Ni SAC was found to be the optimal sample owing to strong CO₂ adsorption and a significant reduction in the rate-determining potential (0.36 eV) required for the formation of formic acid *via* the *OCHO key intermediate. Besides, the findings demonstrate that NiCN is selective towards the CO₂RR despite the inevitable competition from the HER of the water splitting process.

2. Methodology

In this study, all theoretical calculations were performed based on DFT by employing the Vienna *Ab initio* Simulation Package (VASP) with the post-processing VASPKIT package.^{59–63} Generalized gradient approximation with the Perdew–Burke–Ernzerhof (GGA-PBE) exchange–correlation functional was adopted in the calculations. The projected augmented wave (PAW) pseudopotentials and the plane-wave basis functions with a plane wave energy cut-off of 500 eV were used.⁶⁴ A $3 \times 3 \times 1$ Monkhorst–Pack grid *k*-point mesh was used to sample the surface Brillouin zone.⁶⁵ For geometric optimization and ionic self-consistent calculation, the energy convergence criterion was set to 1.0×10^{-4} eV per atom, and force convergence at 0.05 eV Å^{−1} was considered. A $2 \times 2 \times 1$ supercell was constructed with a vacuum space of 15 Å in the *z*-direction to minimize the interlayer interactions. Furthermore, the thermodynamic stability of the selected SAC was studied by *ab initio*

molecular dynamics (AIMD) simulation with a time step of 1.5 fs. Charge distribution in the form of net Bader net atomic charge was calculated using the code provided by the Henkelman group.⁶⁶

The formation energy of the loading of M-SAC on the g-C₃N₄ host material is an important indicator of the stability of M-SAC in the system. The following eqn (1) was employed to define the formation energy (E_f) of the SAC-loaded system:

$$E_f = E_{\text{SAC-CN}} - E_{\text{SAC}} - E_{\text{CN}} \quad (1)$$

where $E_{\text{SAC-CN}}$, E_{SAC} and E_{CN} represent the total energy of the M-SAC-loaded g-C₃N₄ system, the chemical potential of the single metal atom (from the stable metal bulk system), and the energy of the $2 \times 2 \times 1$ g-C₃N₄ monolayer system, respectively.

The Gibbs free energy changes in the reaction coordinate are based on the relative energy changes of the transition state. The Gibbs free energy denotes the energy required to overcome the barrier towards the formation of the next transition state, which directly indicates the likelihood of the formation of the transition state from the energy perspective. Eqn (2) was used to define the adsorption energy (E_{ads}) of the reactant and the energy of the intermediate-product-loaded system:

$$E_{\text{ads}} = E_{\text{total}} - (E_{\text{CN or MCN}} - E_{\text{adsorbate}}) \quad (2)$$

where E_{total} , $E_{\text{CN or MCN}}$ and $E_{\text{adsorbate}}$ represent the total energies of the molecules in the catalytic system, the isolated catalyst and the adsorbate molecule (that is adsorbed or removed from the system), respectively.¹⁵

A screening process (Fig. 1) was developed to screen the optimal M-SAC to be loaded on the g-C₃N₄ system to catalyze the CO₂RR *via* the two-proton coupled electron transfer process. The process consisted of 4 evaluation levels: (1) the thermodynamic and dynamic stability of the M-SAC-loaded system, (2) the CO₂RR activity and product selectivity (the competition between the formation of CO and HCOOH), as well as H₂ evolution reaction activity, (3) the competition between CO₂RR and HER and (4) the optical properties and charge distribution of the M-SAC-loaded system.

3. Results and discussion

3.1 Geometries and the stability of SACs on g-C₃N₄

It is known from the widely reported nitrogen-coordinated metal-single-atom-loaded graphene-based catalysts^{67–71} that the unique tri-*s*-triazine architecture of g-C₃N₄ exhibits a large internal cavity between the triazine structures and the unsaturated nitrogen atoms, which is particularly suitable for the loading of atoms with large atomic radii. As such, this work is focused on loading various M-SACs at the center of the six-fold cavity of g-C₃N₄ and evaluating their role in lowering the rate-determining potential of the CO₂RR. The well-optimized monolayer structure of g-C₃N₄ with C–N bond lengths of around 1.33–1.40 Å and 1.47–1.5 Å in the six-membered ring and the heptazine bridging C–N, respectively, and the lattice size of 7.14 Å ($a = b$) are in good agreement with previous theoretical results.^{72–74} The formation energies of different M-SACs were calculated to evaluate their stability in the inter-

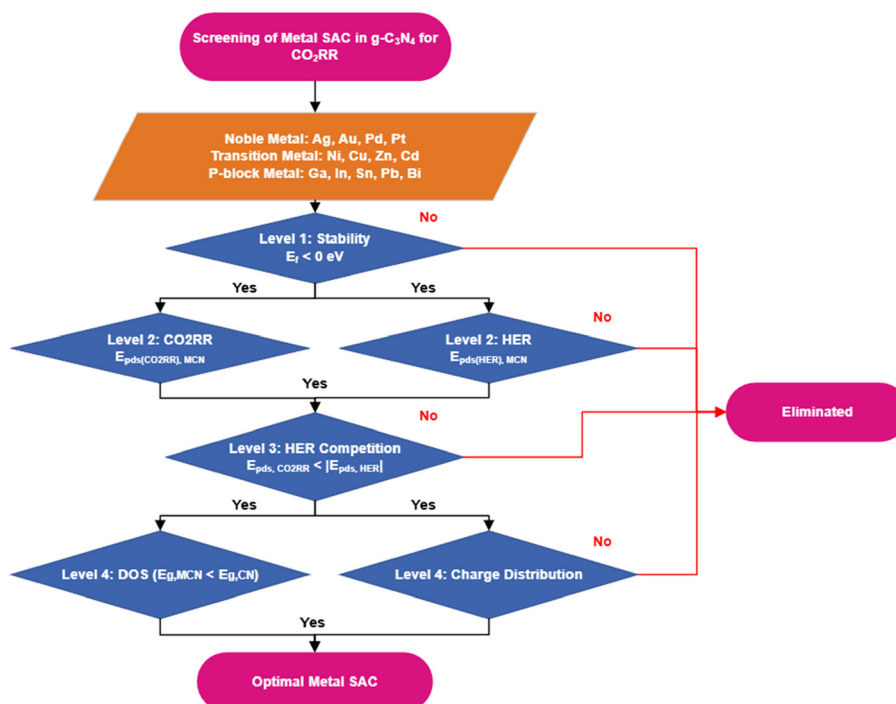


Fig. 1 Process flow chart for screening M-SAC to be loaded on g-C₃N₄ for catalyzing the CO₂RR.

stitial space of the heptazine structure of $g\text{-C}_3\text{N}_4$, as shown in Fig. 2(a). Apart from the Cd and Zn SACs, the formation energies of all other M-SACs had negative values ranging from -0.39 to -2.54 eV. This implies that the loading of the M-SACs is thermodynamically favorable and spontaneous except for the Cd and Zn SACs, as shown in Fig. 2(b). Therefore, the Cd and Zn SACs were excluded from further investigations.

Besides evaluating the thermodynamics of the loading of M-SACs, it is generally known that the N atoms at the edge of the triazine ring serve as the active sites of graphitic carbon nitride due to the unsaturated coordination of the N atoms.⁷⁵ Hence, the interaction between the M atoms and the neighboring unsaturated N atoms plays an important role in retaining the structural integrity and constructing a favorable coordination environment for the activation of CO_2 .⁷⁶ Even though all the metals can be spontaneously loaded onto the interstitial space of the tri-s-triazine allotrope except for Cd (0.57 eV) and Zn (0.85 eV) (Fig. 2(c)), it is expected that elements that exhibit poor interaction with the N atoms are unlikely to exhibit good reduction performance owing to the structural deformation during the activation of the CO_2 molecules (In, Ag, Cd and Ga SACs). On the other hand, the M-SACs with

larger atomic radii, including Bi, Sn, and Pb SACs, were strongly bonded to $g\text{-C}_3\text{N}_4$ with M–N distances ranging from 2.49 – 2.76 Å while residing in the center of the 6-fold cavity. Therefore, the loaded SACs were less likely to aggregate. Meanwhile, the M-SACs with smaller radii (Cu, Ni, Zn, Pd and Pt) tended to shift from the middle of the interstitial space to one of the corners of the heptazine ring, forming bonds with the two neighboring N atoms (M–N distances of 1.89 – 2.23 Å) and resulting in a structure with a lower coordination number.

The thermodynamic stability of an M-SAC is generally represented by the formation energy, and it is correlated with the dynamic stability of the MCN system. Thus, to ensure that the M-SAC has been successfully loaded to the six-fold cavity of the CN system, the Au system with the lowest formation energy was tested. In order to verify the stability of the SAC-loaded CN system, we performed an AIMD simulation of the AuCN system at $T = 300$ K for 4 picoseconds (4 ps). In the simulation, there was no obvious geometry deformation of AuCN (Fig. S1†), and most importantly, the metal atoms did not escape the 6-fold cavity of $g\text{-C}_3\text{N}_4$ throughout the duration of the AIMD simulation, manifesting its stability in the six-fold cavity. This verifies that the interaction between the metal single atom and the

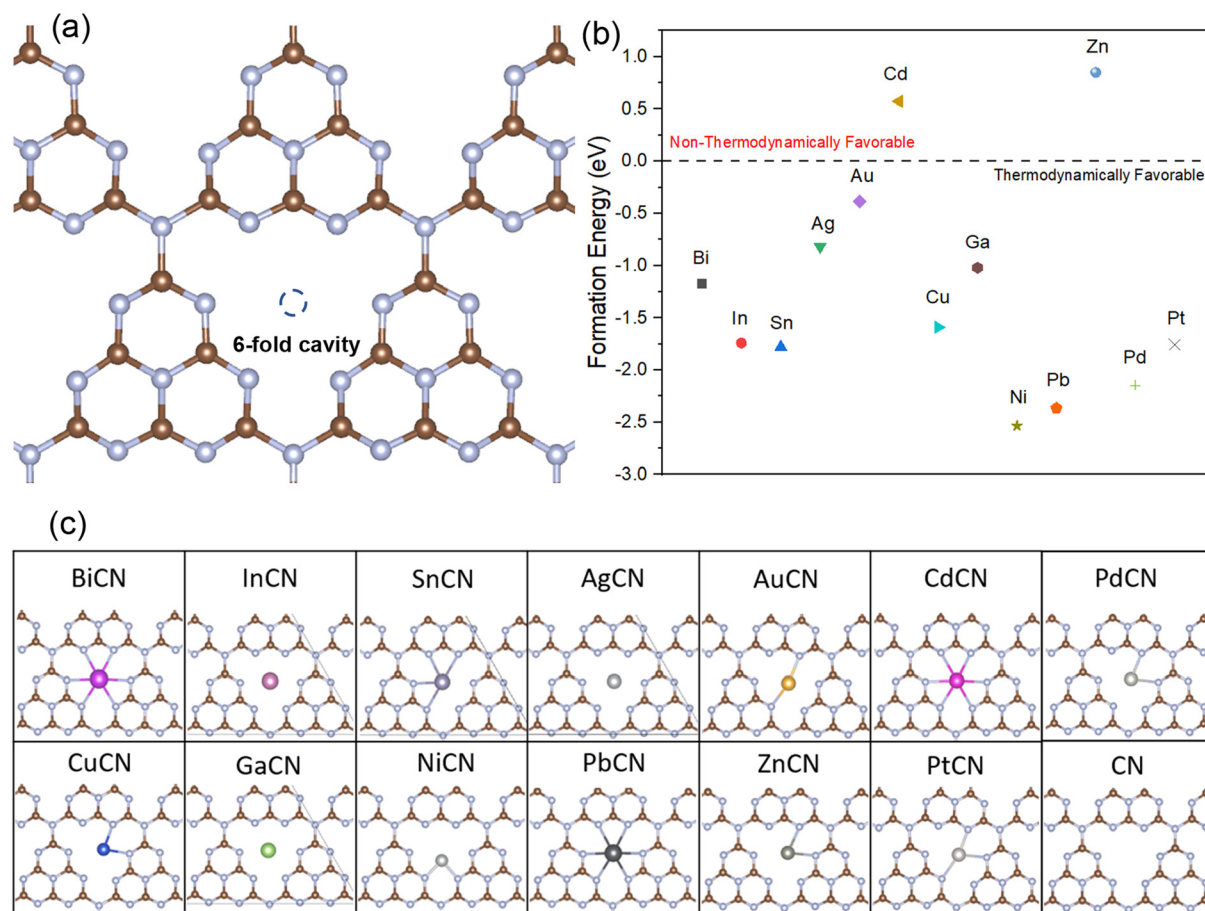


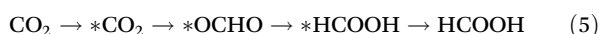
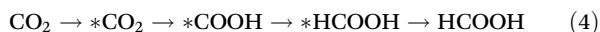
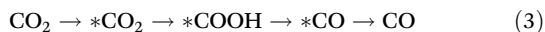
Fig. 2 (a) The optimized corrugated $g\text{-C}_3\text{N}_4$ structures and the M-SAC binding sites of the six-fold cavity. Brown and blue indicate C and N atoms, respectively. (b) The computed formation energy of different M-SACs supported on the six-fold cavity. (c) Optimized structures of the six-fold cavities in the investigated MCN samples.

CN system can stabilize the metal in its atomic form without aggregation.

3.2 CO₂RR pathways

In the first stage of reaction analysis, it is important to identify the relevant reaction pathway of the CO₂RR. The first transfer of the proton–electron pair to CO₂ will lead to the formation of a carboxyl (*COOH) species or formate species (*OCHO), which can be further reduced to various carbonaceous products.⁷⁷ In this study, we particularly investigated the role of SAC in tuning the elementary CO₂RR steps towards governing the product selectivity of the reaction. Therefore, we limited this study to only the 2-proton/electron CO₂RR pathway (*i.e.*, CO₂ to HCOOH and then to CO). The scope of the study was narrowed down to the competition between the formation of CO and HCOOH from CO₂ *via* the carbonate or the formate reaction route.

First, the CO₂ molecule is adsorbed to the surface of the system and is reduced to one of the key intermediates, namely *COOH (carbon-bond configuration) or *OCHO (oxygen-bonded configuration), which initiates the reduction reaction. Subsequently, the second transition state is formed with a further reduction of *COOH or *OCHO to *CO or *HCOOH *via* the 3 reaction pathways given below (eqn (3)–(5)). These are desorbed from the surface, producing CO and HCOOH as the end products of CO₂RR, respectively (Fig. 3).



3.3 CO₂RR activity and selectivity analysis

As mentioned above, only the two-proton coupled electron transfer process was studied in this investigation. The first pro-

tonation of CO₂ is generally known as the potential-limiting transition state, which governs the effectiveness of the system in catalyzing the CO₂RR. The loading of heteroatoms, particularly SAC, is one of the effective methods to alter the catalytic surface chemistry and enhance the catalytic activity of the system by lowering the energy barrier. Hence, noble metals (Ag, Au, Pt and Pd), post-transition metals (Ni, Cu, Zn and Cd) and p-block metals (Ga, In, Sn, Pb and Bi) SAC were loaded onto the surface of g-C₃N₄ and investigated for their effect on tuning the key intermediates to yield different carbon products. The findings suggest that the loading of SACs stabilized the *COOH and *OCHO key intermediates, with all samples showing significantly lower relative free energy changes in comparison with the pristine CN system (Fig. S3(a, b) and Table S2†). Owing to the oxophilicity of p-block metals, all the p-block metals (Ga, In, Sn, Pb and Bi) favored the formation of the O-bonded intermediate *OCHO (over the *COOH intermediate), with the relative Gibbs free energy changes of 0.37, 0.50, −0.82, −0.56 and −0.40 eV, respectively. *OCHO proved to be the key intermediate in the formation of formic acid (Fig. S3(c)†). Unlike transition metals, the post-transition metals possessed p-block-metal-like behavior and mostly showed lower free energy for the formation of the O-bonded intermediate than the C-bonded intermediate.

In order to identify the active reaction sites of the MCN, a few potential active sites were investigated to locate the feasibility of the CO₂RR. The unsaturated edge nitrogen is not only a potential anchoring spot for the M-SAC; it is also a commonly studied catalytically active site in pristine g-C₃N₄. Thus, the CO₂ molecules were first adsorbed on the unsaturated N atoms on g-C₃N₄ and the MCN system. In the pristine CN system, the CO₂ molecules were adsorbed on the saturated N atoms, and thus the edge N atoms on the heptazine framework were selected as the active sites of the CN system. However, the CO₂ reactant is less likely to be adsorbed on the same spots owing to electron redistribution after the loading of the

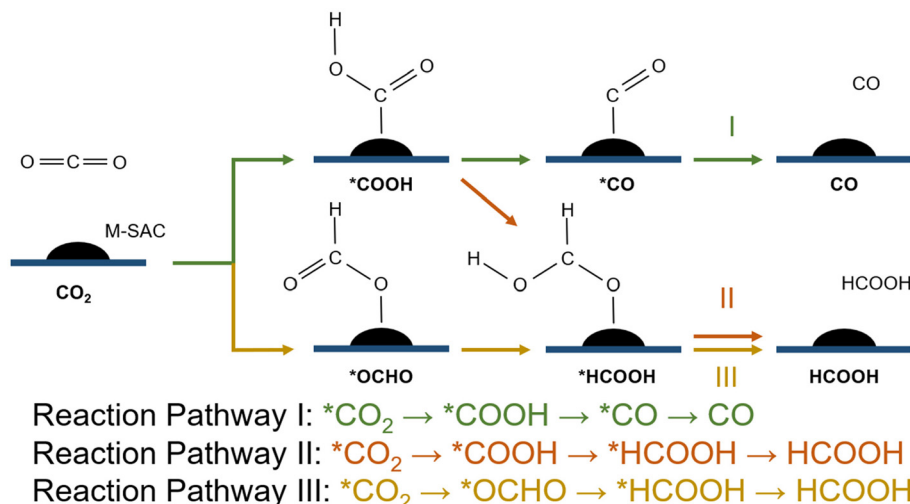


Fig. 3 CO₂RR mechanism. I: CO₂ to CO *via* the *COOH intermediate; II: CO₂ to HCOOH *via* the *COOH intermediate; III: CO₂ to HCOOH *via* the *OCHO intermediate.

M-SAC. Hence, after the adsorption of CO₂ molecules on the M-SAC, it was found that the SAC not only favors the adsorption of CO₂ molecules, but it also significantly lowers the energy barrier by destructing the linear and stable structure of CO₂.

It has been found previously that the lone electron pair on the oxygen of CO₂ tends to interact with the metal single atom, and the molecular behavior is similar to the other M-SAC supported on graphene.⁷⁸ However, the distance between adsorbed CO₂ and the metal single atom (actives) ranged from 1.9 to 3.4 Å due to the weak van der Waals interaction between CO₂ and the M-SAC-loaded carbon nitride surface, which is also reflected in the CN structure (Fig. S2(a)†). A similar phenomenon has been observed by Maihom and group who employed empirical correction in Gimme's scheme (DFT-D3).⁷² Moreover, the Ni SAC, which was shifted to the corner position with lower coordination, exhibited more active CO₂ chemisorption because of its strong electron-donating ability. Evidently, the activation of CO₂ was revealed by the dimensional changes of the CO₂ molecule, with the C–O bond length increasing from 1.17 to 1.21 Å when adsorbed on the surface of the NiCN catalyst.^{79–82} Another apparent evidence of CO₂ activation was observed since the linear angle of CO₂ (180°) was significantly reduced to 143.4° after the geometry optimization of CO₂ adsorbed on the NiCN catalyst (Fig. S2†).

For the formation of CO, in most cases, the rate-determining step is the first hydrogenation step of CO₂, but several exceptions like catalysis on AuCN, CuCN, NiCN, PdCN and PtCN have been reported as they are restricted by the desorption of CO. Hence, it is highly unlikely for the end product of the CO₂RR to be CO, which will be further reduced into other products, such as CH₄, HCOOH, C₂+, and even C₃+, which align with the following discussion on HCOOH.^{14,15,83–86}

From the coordination environment perspective, it was observed that the coordination number of the M-SAC plays a significant role in modulating the potential-limiting steps of the reaction. As mentioned above, due to poor interaction between the M-SAC and neighboring N atoms, M-SACs with lower coordination numbers (InCN, AgCN and GaCN) had a huge increment in the first protonation step of the CO₂ molecule towards forming both *COOH and *OCHO intermediates. This originates from the deformation of the structure due to weak bonding between the M-SAC and the unsaturated N atoms, causing the structures to heavily deform (*i.e.* away from the original plane of the monolayer) and form a bond with the intermediates. On the other hand, the highly coordinated M-SACs (BiCN, CdCN and PbCN) generally had difficulty in the adsorption of the CO₂ molecule, while the *CO desorption process was easy owing to the weak interaction between the SAC and the intermediate. This phenomenon is caused by the dangling and available bonds on the M-SACs that tend to form bonds with the neighboring N atoms. Interestingly, the 6-coordinated M-SACs aid the activation of CO₂ and the formation of *OCHO intermediates. Unlike the M-SACs with high coordination, the low-coordinated SACs (AuCN, CuCN, NiCN and ZnCN) were exquisite in the protonation of CO₂ to form the

*COOH intermediate while exhibiting difficulty in the CO desorption process as a trade-off.

In this analysis, a comparison of the rate-determining potential between the CO₂RR and HER served as an indicator of reaction selectivity; this was evaluated based on the thermodynamic reaction free energy of the CO₂RR step involving the formation of either CO or HCOOH and the reduction of water to hydrogen. Among noble metals, Au SAC showed a significantly low thermodynamic reaction free energy of 0.63 eV, and therefore, <0.63 eV was chosen as the passing criterion for the M-SAC that can substitute the cost-inefficient noble metals. In this stage, there were only 7 metals SACs that met this criterion, namely Bi, In, Cu, Ga, Ni, Pb and Pd SACs. Among them, only 4 SACs exhibited energy barriers less than 0.45 eV, including Ga, Pb and Ni SACs with energy barriers of 0.21, 0.36 and 0.44 eV, respectively. Yet, Ga SAC had poor CO₂ adsorption on the surface, which is higher than any of the transition states in the evolution of CO₂. Thus, it is excluded from property evaluation in the later discussion. Thus, NiCN showed optimal performance among the 13 metal SACs studied. This showcases the potential of Ni SACs supported on carbon nitride as an efficient catalyst for reducing CO₂ to formic acid (Table 1).⁸⁷

3.4 HER competition

The CO₂RR is usually conducted in the liquid–gas phase or the solid–gas (moist CO₂) phase, where the interaction between the catalyst and water is inevitable. Hence, the competition between the HER and CO₂RR is detrimental to the catalytic progression of the CO₂RR. Both HER and CO₂RR will consume the generated h⁺ + e[−] pairs during the protonation process. In addition to that, the H atoms will occupy and cause an active site poisoning effect, demoting the CO₂RR efficiency. Hence, a comparison between the free energy barriers of HER and CO₂RR serves as a great selectivity descriptor in understanding which reaction is more likely to occur on the surface of the MCN system. The reaction with a lower free energy barrier is more favorable than the reaction with a higher free energy barrier owing to the large overpotential requirement in driving the latter reaction (Table 2).

Table 1 Relative free energy of the first protonation of CO₂

Systems	ΔG [*COOH] (eV)	ΔG [*OCHO] (eV)
CN	1.36	2.44
BiCN	0.32	−0.40
InCN	1.24	0.50
SnCN	0.77	−0.82
AgCN	0.73	0.63
AuCN	−0.82	−0.93
CuCN	0.23	−0.13
GaCN	0.87	0.37
NiCN	0.45	0.22
PbCN	0.51	−0.56
PdCN	0.64	0.53
PtCN	−0.45	−0.95

Table 2 Dominant product, PDS and free energy barrier of CO₂RR in the MCN systems based on the lowest rate-determining potential of PDS based on the relative Gibbs free energy changes in completing Pathway 1, 2 or 3 to form CO and HCOOH

Systems	Reaction path	PDS	Potential-limiting energy barrier (eV)
CN	I or II	*CO ₂ → *COOH	1.36
BiCN	I or II	*CO ₂ → *COOH	0.32
InCN	III	*CO ₂ → *OCHO	0.50
SnCN	I or II	*CO ₂ → *COOH	0.77
AgCN	III	*CO ₂ → *OCHO	0.63
AuCN	II	*COOH → HCOOH	1.11
CuCN	II or III	*HCOOH → HCOOH	0.52
GaCN	III	*CO ₂ → *OCHO	0.37
NiCN	III	*OCHO → *HCOOH	0.36
PbCN	III	*OCHO → *HCOOH	0.44
PdCN	III	*CO ₂ → *OCHO	0.53
PtCN	II	*COOH → *HCOOH	0.75

*PDS = potential-determining steps (the transition state that requires the highest potential in the entire reaction path).

By comparing the HER and the two CO₂RR (two-proton/electron pairs) reaction pathways, it was found that there is a strong correlation between the *H and *COOH intermediates, which is similar to the previously reported DFT work.⁸⁸ A similar trend was observed for the formation of CO through the carboxyl (*COOH) intermediate, with the inevitable and strong HER competition causing the formation of CO to be less competitive to the formation of H₂. A majority of CO-selective MCN systems (BiCN, SnCN, CdCN, ZnCN) had strong HER competition, as exemplified by the much lower free energy barrier of the HER reaction. A similar phenomenon was found in the noble metal systems AuCN, PdCN and PtCN, which had rate-determining potentials of 0.87, 0.39 and 0.28 eV, respectively. On the other hand, the CO₂RR pathway with the formate intermediate (*OCHO) was found to retain selectivity towards the CO₂RR over the HER (Table 3).

Table 3 Rate-determining potentials of the CO, HCOOH formation reactions and HER on the CN and MCN systems and their selectivity to CO₂RR or HER. The values in the table correspond to Fig. 4

Systems	ΔG_{pds} HER (eV)	ΔG_{pds} CO ₂ RR (eV)	Selectivity
CN	1.80	1.36	CO ₂ RR
BiCN	0.18	0.31	HER
InCN	1.02	0.50	CO ₂ RR
SnCN	0.51	0.77	HER
AgCN	0.67	0.63	CO ₂ RR
AuCN	0.87	1.11	HER
CdCN	0.16	0.32	HER
CuCN	0.42	0.52	HER
GaCN	0.88	0.37	CO ₂ RR
NiCN	0.75	0.36	CO ₂ RR
PbCN	0.59	0.44	CO ₂ RR
ZnCN	0.43	0.45	HER
PdCN	0.39	0.53	HER
PtCN	0.28	0.75	HER

*HER = Hydrogen evolution reaction, CO₂RR = Carbon dioxide reduction to carbon monoxide *via* *COOH as the key intermediate or to formic acid *via* *OCHO as the key intermediate (depending on which has a lower free energy barrier).

3.5 Property investigation and charge distribution

For an emerging photocatalytic material, the bandgap indicates its capability to harvest light and the absorption range of the light to induce the formation of electron-hole pairs for the photocatalytic redox reaction. The bandgap of g-C₃N₄ was calculated to be 1.85 eV, which is way below the experimental value, when the PBE functional was formulated to obtain the ground-state properties of atoms, molecules and condensed matter. However, it could not predict the exact bandgaps of semiconductors. The fundamental bandgap of a system with N electrons is given by the differences in the total ground-state energies of systems with different numbers of electrons. The ground-state total energy can be estimated by DFT but this does not imply that the bandgap of the Kohn-Sham system is the fundamental gap of the interacting electron system. Correction using HSE03 can be implemented to rectify the

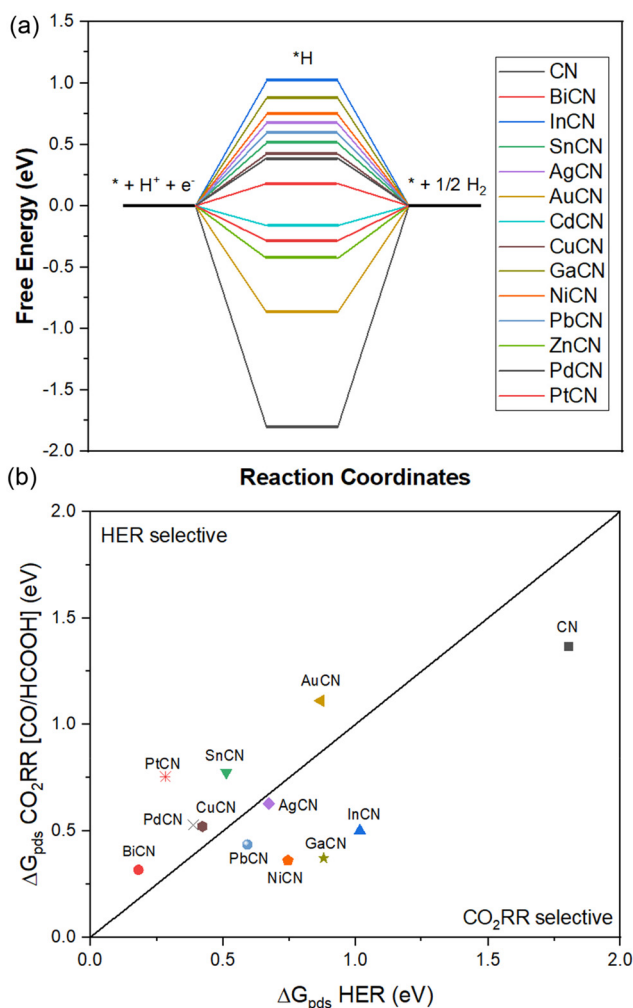


Fig. 4 (a) Free energy diagram of HER on the SAC-loaded CN systems (M = Bi, In, Sn, Ag, Au, Cd, Cu, Ga, Ni, Pb and Zn). (b) Competition between CO₂RR and HER based on the rate-determining potentials of the CN and MCN systems. The area below the linear lines indicates CO₂RR-selective catalysts, whereas the area above the linear line indicates HER-selective catalysts.

underestimation of the bandgap, but it requires a lot of computational resources and is less suitable for calculating a $2 \times 2 \times 1$ supercell to compare the pristine and the metal-loaded systems. Yet, the PBE generated can serve well to compare the relative bandgap changes between pristine CN and the metal-loaded counterpart. In our analysis, the N 2p orbital dominated the valence band maximum (VBM), while the conduction band minimum (CBM) was mainly composed of a combination of the 2p orbitals of C and N atoms, similar to that reported in the literature.⁷² Two M-SACs, namely Pb and Ni SACs, had narrower bandgaps, indicating that the metal-

loaded systems have better light adsorption capability than pristine CN (Fig. 5). Furthermore, it was observed that the mid-gap state, which is also known as the impurities state of the metal-loaded SAC, was located closer to the CB. As such, the impurity states will serve as the electron capture sites and thus active sites for the reduction process, once again verifying the better adsorption of the reactant and other molecules to the M-SAC. Besides, the loading of M-SAC was found to have a band-shifting effect that governs the reducibility of the catalyst system. All three M-SAC-loaded CN samples had upshifted peaks, indicating the stronger reduction potential of the

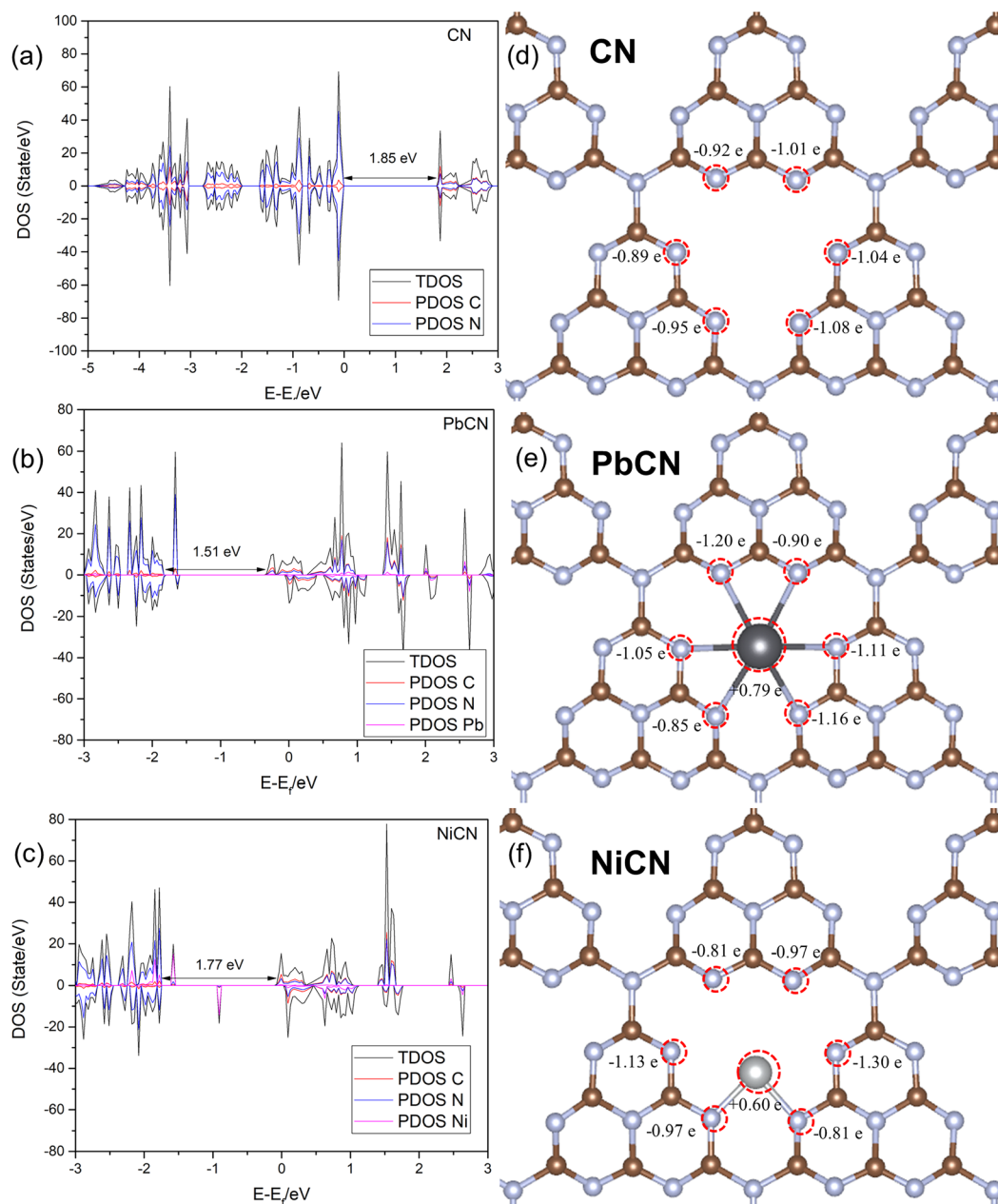


Fig. 5 The density of states of the metal-loaded CN systems: (a) pristine CN, (b) PbCN and (c) NiCN. Bader net atomic charges on the catalytic systems: (d) pristine CN, (e) PbCN and (f) NiCN. The brown balls represent carbon atoms; the light blue balls represent nitrogen atoms; the black ball represents the Pb atom and the silvery grey ball represents the Ni atom.

systems. The two M-SAC-loaded systems PbCN and NiCN appeared to have strong light harvesting ability with lowered bandgaps of 1.51 and 1.71 eV, respectively.

Bader charge analysis was conducted to investigate the charge distribution on the catalyst surface to elucidate the charge transfer process after the loading of the M-SACs. It was observed that the loading of different metal-SACs leads to different charge redistribution on the catalyst and that the neighboring N atoms contribute electrons to the M-SACs, which is in coherence with the discussion on the density of states of the system that the M-SAC serves as the electron-trapping zone. Both the systems showed charge migration effect on the Pb and Ni SAC sites with positive charges at +0.79 $|e|$ and +0.60 $|e|$, respectively, while the total charge changes in the surrounding N atoms were $\pm 0.38 |e|$ and $\pm 0.10 |e|$ in comparison with the pristine CN system (Fig. 5). This reflects that charges are contributed by the neighboring N atoms to the loaded M-SAC, easing the adsorption of the CO₂ molecule, as reflected by the lowered energy barrier in the previous discussion. Electron transfer from the metal atoms to g-C₃N₄ leads to a more positive charge value on the metal single atoms, once again confirming the strong covalent interactions between the metal single atoms and g-C₃N₄.

4. Conclusion

In summary, we applied periodic DFT calculations to investigate how CO₂ is adsorbed and reduced to formic acid in the MCN systems. The metal active sites attract CO₂ through the electron withdrawal from the neighbouring N atoms and accumulation on the metal active sites. From the reaction kinetics perspective, NiCN showed outstanding CO₂ activation ability, as observed from the bent structure of CO₂ from 180° to 143.3° and the lengthened C–O bond from 1.17 to 1.21 Å after adsorption on the catalyst surface. This is mainly because of the electrons that migrate from the neighboring N atoms to the Ni atoms (0.60 $|e|$), which drastically improve the adsorption and activation of the CO₂ molecule on the surface. By comparing the potential-determining steps across various pathways leading to the formation of carbon monoxide and formic acid, it was found that NiCN shows optimal performance with the lowest Gibbs Free reaction energy change of 0.36 eV, which is approximately a 73.5% improvement compared with pristine CN. Moreover, NiCN is also selective towards the formation of formic acid, with a significantly lowered rate-determining potential of <0.5 eV; It selectively converts CO₂ to formic acid in two steps (Pathway 3): (i) CO₂ is first hydrogenated to the formate intermediate state; (ii) the formate intermediate is then further reduced to formic acid. The hydrogenation of the formate intermediate to formic acid is the potential-determining step of the NiCN system. The lower activation energy of the hydrogenation steps in NiCN is due to the greater electron delocalization between the catalyst active site and the C–O bond of CO₂, which leads to a more stable transition state. Apart from that, the free energy change of the CO₂RR potential-determining step to gene-

rate formic acid (0.36 eV) is lower than that of the highly competitive HER (0.75 eV), thus inferring that the catalytic system favors CO₂RR over the HER. In conclusion, our results have established an inspiring framework based on CO₂ adsorption and hydrogenation to produce formic acid on graphitic carbon nitride with metal-single-atom modifications for advancing the design of future catalysts.

Data availability

ESI† associated with this article can be found, in the online version, at doi: 10.1016/

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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